

Problem 4, Cell 1: Sun's conjecture for $k = 3$

Official statement. For integers a and $m \geq 1$ write $a \pmod{m} = \{a + mt : t \in \mathbb{Z}\}$. A finite family of classes $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ is pairwise disjoint if $a_i \pmod{m_i} \cap a_j \pmod{m_j} = \emptyset$ whenever $i < j$. Nothing is assumed about the moduli beyond $m_i \geq 1$: they may repeat, and the residues a_i may repeat as well. By the Chinese remainder theorem,

$$a_i \pmod{m_i} \cap a_j \pmod{m_j} \neq \emptyset \iff \gcd(m_i, m_j) \mid a_i - a_j. \quad (*)$$

Question. Let $k \geq 2$ and let $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ be pairwise disjoint. Must there be a pair $i < j$ with $\gcd(m_i, m_j) \geq k$?

The first size that the two trivial observations above do not settle. Prove the statement for $k = 3$: any three pairwise disjoint classes have $\gcd(m_i, m_j) \geq 3$ for some $i < j$.

Theorem 1 below is exactly this statement. The notation $a \pmod{m}$ below is the class $a \pmod{m}$ of the statement. Lemma 2 is $(*)$, proved for completeness.

Known result. The question is Sun's disjoint congruence classes conjecture [2]. O'Bryant [1] proved it for all $k \leq 20$, so this case is known. The proof below is an independent proof and is self-contained.

For an integer $m \geq 1$ and $a \in \mathbb{Z}$ write $a \pmod{m} = \{x \in \mathbb{Z} : x \equiv a \pmod{m}\}$. A family of residue classes is *disjoint* if its members are pairwise disjoint.

Theorem 1. *Let $a_1 \pmod{m_1}$, $a_2 \pmod{m_2}$, $a_3 \pmod{m_3}$ be three pairwise disjoint residue classes. Then $\gcd(m_i, m_j) \geq 3$ for some $i \neq j$.*

1 When do two classes meet?

Lemma 2. *Let $m, n \geq 1$, $a, b \in \mathbb{Z}$ and $g = \gcd(m, n)$. Then $(a \pmod{m}) \cap (b \pmod{n}) \neq \emptyset$ if and only if $g \mid a - b$.*

Proof. $(\Rightarrow)$ If $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$, then $g \mid m$ and $g \mid n$ give $a \equiv x \equiv b \pmod{g}$.

$(\Leftarrow)$ By Bézout there are $u, v \in \mathbb{Z}$ with $um + vn = g$. Write $a - b = gt$ with $t \in \mathbb{Z}$ and set $x = a - umt$. Clearly $x \equiv a \pmod{m}$. Also $x = a - (g - vn)t = a - gt + vnt = b + vnt \equiv b \pmod{n}$. $\square$

Corollary 3. *Let $a \pmod{m}$ and $b \pmod{n}$ be disjoint and $g = \gcd(m, n)$. Then $g \geq 2$ and $a \not\equiv b \pmod{g}$. In particular, if $g = 2$, then $a \not\equiv b \pmod{2}$.*

Proof. By Lemma 2, disjointness means $g \nmid a - b$. Since 1 divides every integer, $g \neq 1$. $\square$

2 Proof of Theorem 1

Suppose, for contradiction, that $\gcd(m_i, m_j) \leq 2$ for all $i \neq j$. By Corollary 3 each of these gcds is at least 2, so $\gcd(m_i, m_j) = 2$ for all three pairs. By Corollary 3 again, $a_i \not\equiv a_j \pmod{2}$ for all $i \neq j$. In other words, a_1, a_2, a_3 lie in three distinct residue classes modulo 2. There are only two such classes, so this is impossible by the pigeonhole principle. Hence some pair satisfies $\gcd(m_i, m_j) \geq 3$. $\square$

Remark 4. The bound is sharp: $0 \bmod 3$, $1 \bmod 3$, $2 \bmod 3$ are disjoint and every pairwise gcd equals 3. The case of a modulus $m_i = 1$ needs no separate treatment: then $a_i \bmod m_i = \mathbb{Z}$, which meets every class, and the corresponding gcds equal 1, which Corollary 3 excludes.

References

- [1] K. O'Bryant, *On Z.-W. Sun's disjoint congruence classes conjecture*, in *Combinatorial Number Theory*, de Gruyter, Berlin, 2007, pp. 403–411; arXiv:math/0604347.
- [2] Z.-W. Sun, *Finite covers of groups by cosets or subgroups*, *Internat. J. Math.* **17** (2006), 1047–1064; arXiv:math/0501451.